

TIC 411434067

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 1.353604 d (ALIASED, see below)
-> corrected period: 2.707209 d
Epoch (t0) 2825.8292 BTJD
Duration 0.1200 d
Depth 4.289 % (42,887 ppm)
SNR 192.9

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

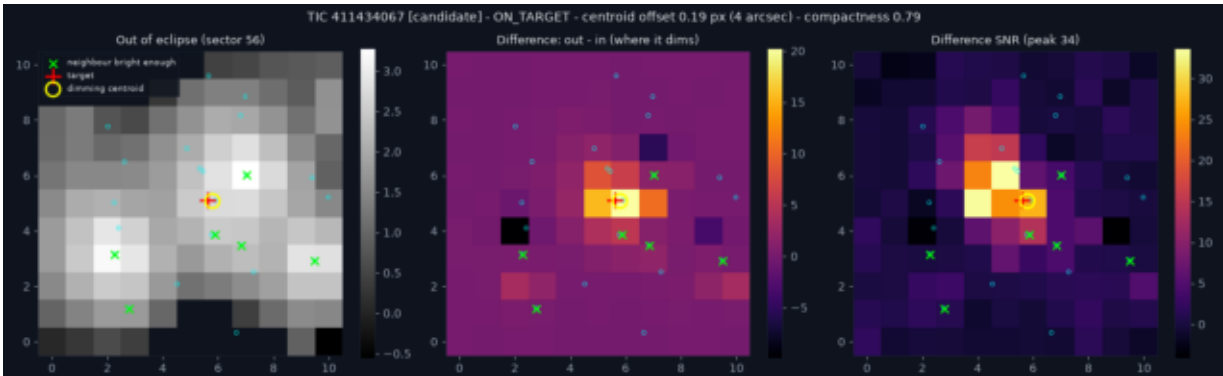
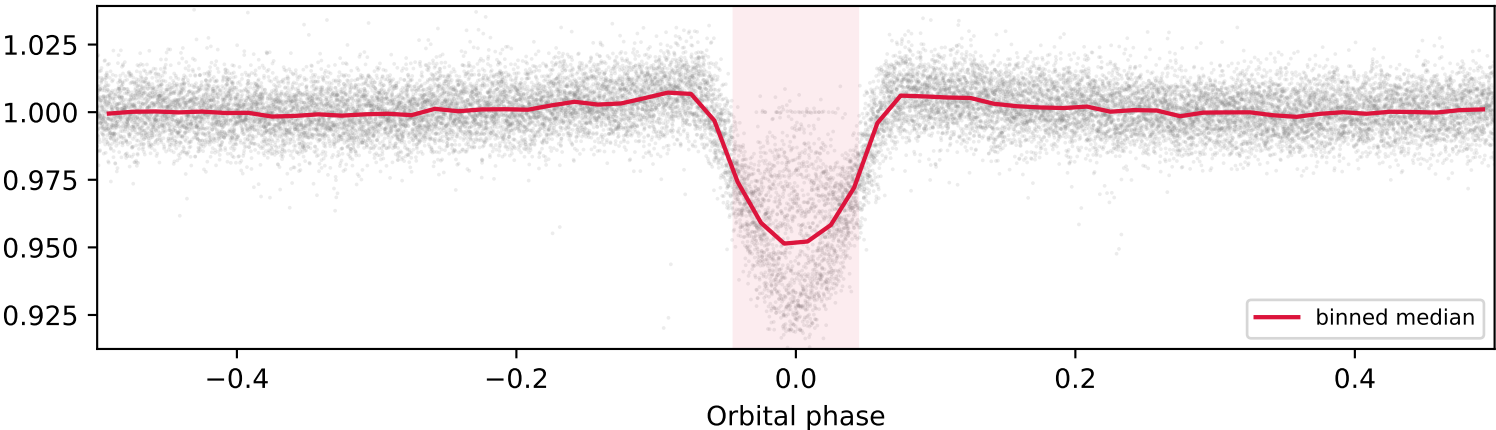
Eclipses observed 39 (gate >= 3)
Robust depth ratio 0.92 (gate >= 0.50)
Robust depth sigma 162.8 (gate >= 7.0)
Odd/even z-score 9.80 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 56 light-curve dip in this sector: 130.8 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 33.70 (gate >= 5.0) Compactness 0.79 (gate >= 0.50)
Dimming centroid offset 0.19 px (4 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 411429819
1.21 px away, dTmag +1.19 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 9.80 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 2.707209 d, in any write-up.